

Hypothesis Test Notes

Business Objective

The objective of this experiment is to determine whether the new onboarding and activation campaign (Treatment group) performs better than the existing onboarding experience (Control group) in improving user conversion while maintaining acceptable business performance.

Primary Metric

North Star Metric: Paid Conversion Rate

Reason for Selection:

Paid Conversion Rate directly measures the percentage of users who become paying customers after experiencing the onboarding process. Improving this metric directly contributes to business revenue and long-term subscription growth.

Null Hypothesis (H_0)

There is **no statistically significant difference** in the Paid Conversion Rate between the Control group and the Treatment group.

$$H_0 : p_{control} = p_{treatment}$$

Alternative Hypothesis (H_1)

The Treatment group has a **higher Paid Conversion Rate** than the Control group.

$$H_1 : p_{treatment} > p_{control}$$

Type of Test

One-tailed A/B Test

Reason:

The business objective is specifically to determine whether the new onboarding experience improves user conversion, not merely whether it is different.

Significance Level

$\alpha = 0.05$ (5%)

Decision confidence = 95%

Test Inputs

- Control Group Users
 - Treatment Group Users
 - Number of Paid Conversions
 - Paid Conversion Rate
 - Sample Size
 - Standard Error
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Statistical Method

A Two-Proportion Z-Test (One-tailed) was selected because:

- Two independent groups are being compared.
 - The outcome variable (Paid Conversion) is binary (Converted / Not Converted).
 - Sample sizes are sufficiently large for normal approximation.
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Decision Rule

- If **P-value** < **0.05**
 - Reject the Null Hypothesis.
 - Conclude that the Treatment significantly improves Paid Conversion Rate.
 - If **P-value** $\geq$ **0.05**
 - Fail to reject the Null Hypothesis.
 - There is insufficient evidence that the Treatment improves conversion.
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Guardrail Metrics Evaluated

The experiment should not be judged only on conversion improvement. The following guardrail metrics were also considered:

- Refund Rate

- Support Ticket Rate
- Average Engagement Score
- Average Days to Convert
- Revenue Quality

These metrics help ensure that any increase in conversions does not negatively affect customer experience or business quality.

Business Interpretation

If the Treatment group demonstrates a statistically significant improvement in Paid Conversion Rate while maintaining acceptable guardrail metrics, the company should consider launching the new onboarding experience.

If conversion improves but guardrail metrics deteriorate significantly (for example, higher refund requests or increased support tickets), additional testing or partial rollout should be recommended before a full launch.

Assumptions

- Users were randomly assigned to Control and Treatment groups.
 - Observations are independent.
 - Data accurately represents user behavior during the experiment.
 - Sample size is sufficient for hypothesis testing.
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Limitations

- Results apply only to the observed experiment period.
 - External factors (seasonality, marketing campaigns, or promotions) may influence user behavior.
 - Segment-level differences may require additional analysis before a full rollout recommendation.
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Conclusion

The hypothesis test provides statistical evidence to support or reject the launch of the new onboarding campaign. The final recommendation should consider both statistical significance and business guardrail metrics before deciding whether to launch the Treatment experience to all users.